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TUGAS PERTEMUAN 3

PERCABANGAN, PERULANGAN, DAN OPERASI GIT LANJUTAN PADA DART MOBILE PROGRAMMING

Link Github: https://github.com/Ailsasahda13/semester4-PemrogramanMobile-TugasJobsheet3

1. Buat program Dart sederhana yang menggunakan kombinasi percabangan dan perulangan untuk menyelesaikan masalah. Contoh: program untuk menghitung faktorial, cek bilangan prima, atau game tebak angka

   -CODE GAME TEBAK ANGKA-

   - Branch Initial-setup

```
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git checkout initial-setup
Switched to branch 'initial-setup'
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git add .
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git commit -m "inisialisasi
variabel angka rahasia"
[initial-setup a7f94a5] inisialisasi variabel angka rahasia
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git push origin initial-setu
p
Enumerating objects: 24, done.
Counting objects: 100% (24/24), done.
Delta compression using up to 16 threads
Compressing objects: 100% (18/18), done.
Writing objects: 100% (23/23), 5.63 KiB | 1.13 MiB/s, done.
Total 23 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
remote:
remote: Create a pull request for 'initial-setup' on GitHub by visiting:
remote:      https://github.com/Ailsasahda13/semester4-PemrogramanMobile-TugasJobsheet3/pull/ne
w/initial-setup
remote:
To https://github.com/Ailsasahda13/semester4-PemrogramanMobile-TugasJobsheet3.git
 * [new branch]      initial-setup -> initial-setup
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git checkout logic-validatio
n
```

   Commit ke-1

```
TugasJobsheet3 > tugasjobsheet3 > bin > main.dart > main
Run | Debug
void main() {
  int angkaRahasia = 136;
  print("Selamat datang di Game Tebak Angka!");
  print("Coba tebak angka rahasianya berapa ya kira kiraa?");
}
```

   - Branch logic-validation

```
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git checkout logic-validatio
n
Switched to branch 'logic-validation'
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git add .
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git commit -m "menambahkan l
ogika percabangan pengecekan angka"
[logic-validation 7053dae] menambahkan logika percabangan pengecekan angka
 1 file changed, 4 insertions(+), 4 deletions(-)
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git add .
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git commit -m "menambahkan v
alidasi input angka positif"
[logic-validation 95619a0] menambahkan validasi input angka positif
 1 file changed, 3 insertions(+)
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git push origin logic-valida
tion
Enumerating objects: 22, done.
Counting objects: 100% (22/22), done.
Delta compression using up to 16 threads
Compressing objects: 100% (15/15), done.
Writing objects: 100% (21/21), 3.74 KiB | 1.25 MiB/s, done.
Total 21 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
remote:
remote: Create a pull request for 'logic-validation' on GitHub by visiting:
remote:      https://github.com/Ailsasahda13/semester4-PemrogramanMobile-TugasJobsheet3/pull/ne
w/logic-validation
remote:
To https://github.com/Ailsasahda13/semester4-PemrogramanMobile-TugasJobsheet3.git
 * [new branch]      logic-validation -> logic-validation
```

Commit ke-2

```
TugasJobsheet3 > tugasjobsheet3 > bin > main.dart > main
Run | Debug
void main() {
  int angkaRahasia = 136;
  int tebakan = 150;

  if (tebakan > angkaRahasia) {
    print("Yeayy, tebakanmu tepat nihh!!!");
  } else if (tebakan < angkaRahasia) {
    print("Hmm, Sorry tebakannya masi salah nihh!!!");
  } else {
    print("Congratulationss! Tebakanmu tepat 136.");
  }
}
```

Commit ke-3

```
TugasJobsheet3 > tugasjobsheet3 > bin > main.dart > main
Run | Debug
void main() {
  int angkaRahasia = 136;
  int tebakan = 150;

  if (tebakan <= 0) {
    print("Error: Masukkan angka positif!");
  }
  if (tebakan > angkaRahasia) {
    print("Yeayy, tebakanmu tepat nihh!!!");
  } else if (tebakan < angkaRahasia) {
    print("Hmm, Sorry tebakannya masi salah nihh!!!");
  } else {
    print("Congratulationss! Tebakanmu tepat 136.");
  }
}
```

- Branch loop-implementattion

```
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git checkout loop-implementation
Switched to branch 'loop-implementation'
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git add .
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git commit -m "implementasi perulangan while"
[loop-implementation 85dd762] implementasi perulangan while
 1 file changed, 34 insertions(+)
 create mode 100644 TugasJobsheet3/tugasjobsheet3/bin/main.dart
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git add .
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git commit -m "finalisasi ga
me dengan penghitung percobaan"
[loop-implementation 94d0050] finalisasi game dengan penghitung percobaan
 1 file changed, 8 insertions(+), 17 deletions(-)
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git push origin loop-implementation
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
 1 file changed, 8 insertions(+), 17 deletions(-)
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git push origin loop-implementation
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
ntation
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Compressing objects: 100% (6/6), done.
Writing objects: 100% (12/12), 1.37 KiB | 155.00 KiB/s, done.
Total 12 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'loop-implementation' on GitHub by visiting:
remote:      https://github.com/Ailsasahda13/semester4-PemrogramanMobile-TugasJobsheet3/pull/new/loop-implementation
remote:
To https://github.com/Ailsasahda13/semester4-PemrogramanMobile-TugasJobsheet3.git
 * [new branch]      loop-implementation -> loop-implementation
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3>
```

Commit ke-4

TugasJobsheet3 > tugasjobsheet3 > bin > main.dart > main

```dart
import 'dart:io';

Run | Debug
void main() {
  int angkaRahasia = 136;
  int? tebakan;
  int percobaan = 0;

  print("=== GAME TEBAK ANGKA (1-200) ===");

  while (tebakan != angkaRahasia) {
    stdout.write("Masukkan tebakanmu: ");

    String? input = stdin.readLineSync();
    tebakan = int.tryParse(input ?? "");

    if (tebakan == null) {
      print("Error: Masukkan angka yang valid!");
      continue;
    }

    percobaan++;

    if (tebakan <= 0) {
      print("Error: Masukkan angka positif!");
    } else if (tebakan > angkaRahasia) {
      print("Hmm, Tebakanmu terlalu TINGGI nihh!!!");
    } else if (tebakan < angkaRahasia) {
      print("Hmm, Tebakanmu terlalu RENDAH nihh!!!");
    } else {
      print("Congratulationss! Tebakanmu tepat 136.");
      print("Kamu berhasil menebak dalam $percobaan kali percobaan.");
    }
  }
}
```

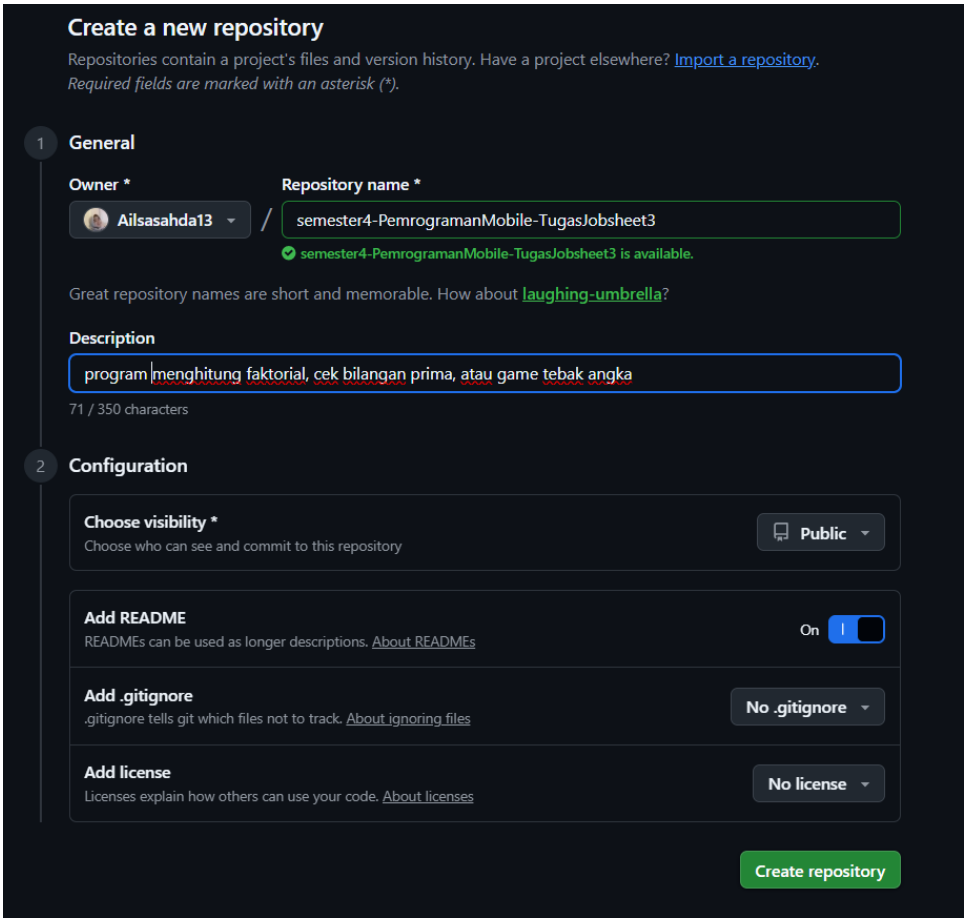
Commit ke-5

```
TugasJobsheet3 > tugasjobsheet3 > bin > main.dart > main
import 'dart:io';

Run | Debug
void main() {
  int angkaRahasia = 136;
  int? tebakan;
  int percobaan = 0;

  print("=== GAME TEBAK ANGKA (1-200) ===");

  while (tebakan != angkaRahasia) {
    stdout.write("Masukkan tebakanmu: ");

    String? input = stdin.readLineSync();
    tebakan = int.tryParse(input ?? "");

    if (tebakan == null) {
      print("Error: Masukkan angka yang valid!");
      continue;
    }

    percobaan++;

    if (tebakan <= 0) {
      print("Error: Masukkan angka positif!");
    } else if (tebakan > angkaRahasia) {
      print("Hmm, Tebakanmu terlalu TINGGI nihh!!!");
    } else if (tebakan < angkaRahasia) {
      print("Hmm, Tebakanmu terlalu RENDAH nihh!!!");
    } else {
      print("Congratulationss! Tebakanmu tepat 136.");
      print("Kamu berhasil menebak dalam $percobaan kali percobaan.");
    }
  }
}
```

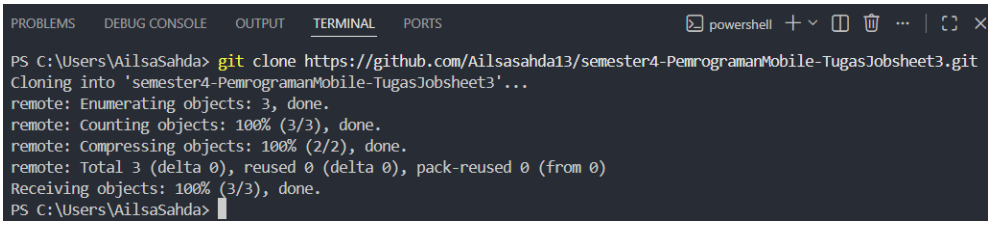
- Hasil

```
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> dart TugasJobsheet3/tugasjobsheet3/bin/main.dart
=== GAME TEBAK ANGKA (1-200) ===
Masukkan tebakanmu: -23
Error: Masukkan angka positif!
Masukkan tebakanmu: 70
Hmm, Tebakanmu terlalu RENDAH nihh!!!
Masukkan tebakanmu: 100
Hmm, Tebakanmu terlalu RENDAH nihh!!!
Masukkan tebakanmu: 120
Hmm, Tebakanmu terlalu RENDAH nihh!!!
Masukkan tebakanmu: 136
Congratulationss! Tebakanmu tepat 136.
Kamu berhasil menebak dalam 5 kali percobaan.
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3>
```

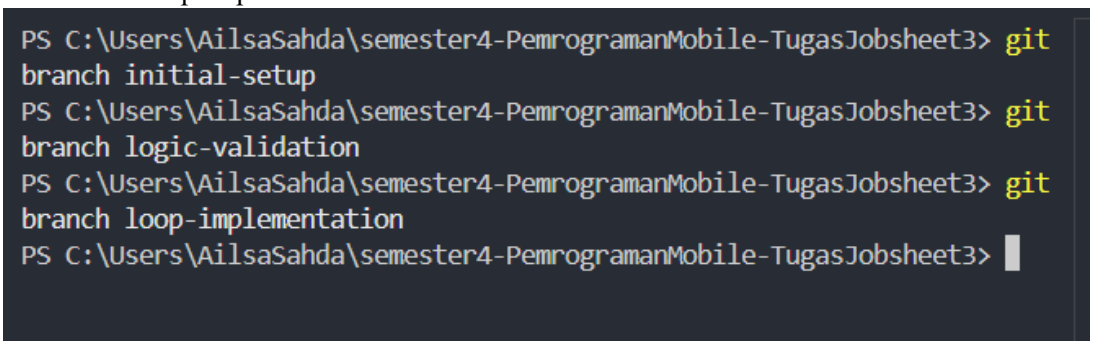
2. Buat repository GitHub untuk program tersebut dengan minimal 3 branch dan 5 commit yang menunjukkan perkembangan fitur
   - Membuat Repository

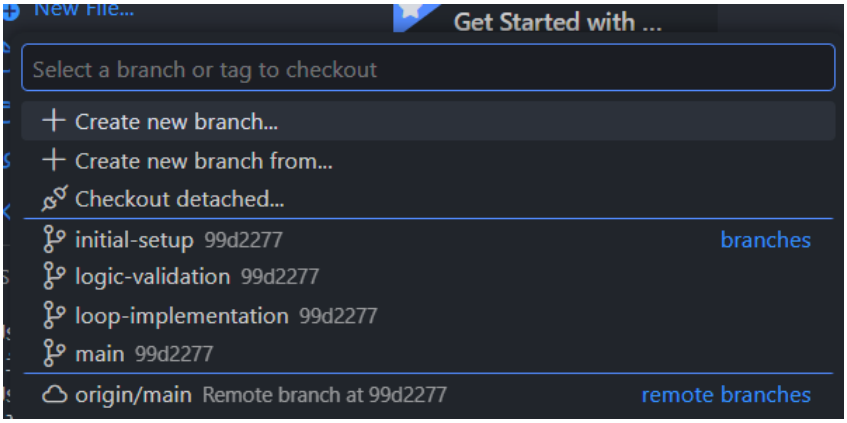


- Clone

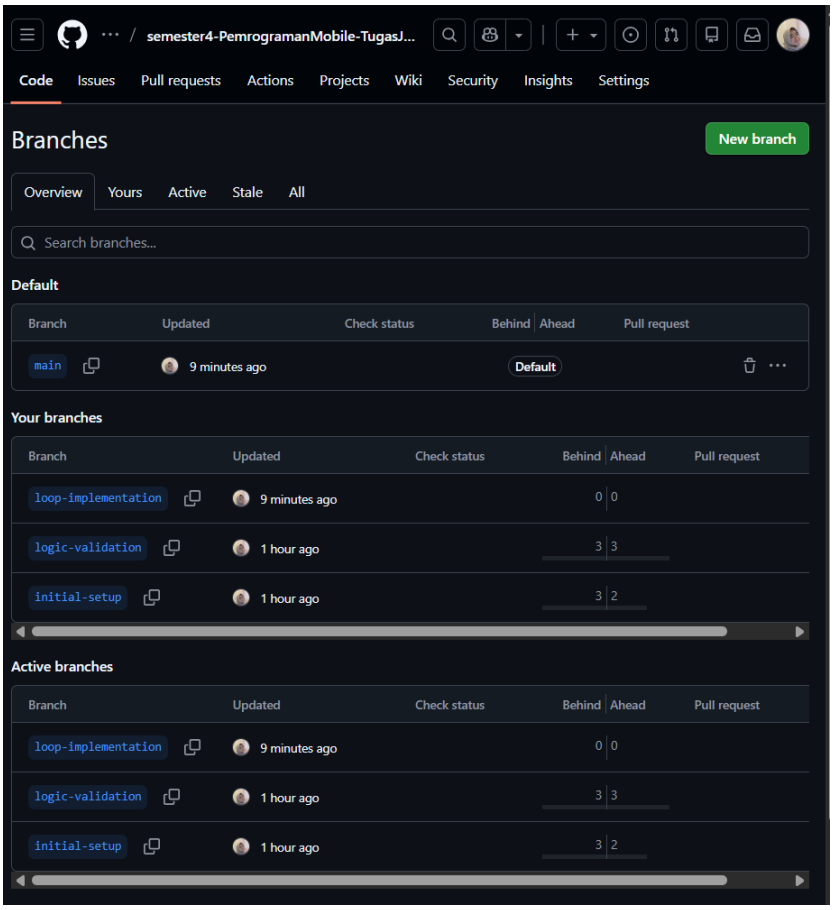


- Branch
  1. Branch initial-setup
  2. Branch logic-validation
  3. Branch loop-implementation





- Branch



- Push ke Main

```
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git merge loop-implementation
Updating 99d2277..bf8310e
Fast-forward
 TugasJobsheet3/tugasjobsheet3/bin/main.dart | 34 ++++++++++++++++++++++++++++++
 1 file changed, 34 insertions(+)
 create mode 100644 TugasJobsheet3/tugasjobsheet3/bin/main.dart
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Ailsasahda13/semester4-PemrogramanMobile-TugasJobsheet3.git
   99d2277..bf8310e  main -> main
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-TugasJobsheet3>
```